

Gandharva Appagere

EXPERIMENTAL PHYSICIST • DETECTORS, DATA & AI

PhD candidate in experimental nuclear & hadron physics, Stockholm University

Stockholm

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PROFILE

PhD candidate in experimental nuclear and hadron physics at Stockholm University, studying hyperon–hyperon ($\Lambda\Lambda$) interactions with the HADES experiment at GSI and simulation-based feasibility studies for the CBM experiment within the QCD at FAIR programme.

EDUCATION

2021–2027 (expected)	PhD, Experimental Nuclear & Hadron Physics <i>Stockholm University, Department of Physics (Fysikum)</i> Stockholm, Sweden Thesis: “Hyperon–hyperon interactions from proton–proton collisions with HADES and CBM” Supervisors: Prof. Per-Erik Tegnér (main supervisor since Dec 2024); Dr. María Doncel Monasterio (main supervisor 2021–2024, now co-supervisor); Dr. Jenny Taylor, GSI (co-supervisor since 2025); Dr. Michael Papenbrock, Uppsala University (co-supervisor 2022–2025). Collaborations HADES and CBM at GSI/FAIR (Darmstadt, Germany); QCD at FAIR programme.
2021–2024	Licentiate of Philosophy (Fil.lic.), Physics <i>Stockholm University</i> Stockholm, Sweden Thesis: “ Λ - Λ hyperons: the road to estimate the production cross-section in proton-proton collisions” Licentiate seminar 27 March 2024.
2018–2020	MSc, Electromagnetics, Fusion and Space Engineering <i>KTH Royal Institute of Technology</i> Stockholm, Sweden Master’s thesis: “Can FIMPs be Dark Matter Candidates?” Thesis advisor: Prof. Robert Mann (Perimeter Institute / University of Waterloo). Thesis on symmetry breaking with the Coleman–Weinberg effective potential at finite temperature.
2013–2017	BE, Electronics and Communication Engineering <i>BMS College of Engineering</i> Bengaluru, India Thesis: “CLAIRE: Cognitive Linguistic Analyser and Interpreter with Real-time Efficiency”

RESEARCH EXPERIENCE

Aug 2021 – present	PhD Candidate (Doktorand) <i>Stockholm University • HADES & CBM collaborations, GSI/FAIR</i> Stockholm, Sweden & Darmstadt, Germany Experimental hadron physics on the HADES spectrometer at GSI and simulation studies for the CBM experiment at FAIR. • Main analysis responsibility in the HADES collaboration for the $\Lambda\Lambda$ channel in proton–proton collisions at $\sqrt{s} = 3.46$ GeV. • Took detector shifts during the five-week February 2022 proton beam time and a 2025 pion-beam run; worked with drift-chamber online quality assurance. • Developed a relative time-of-flight particle-identification method that compensates start-detector timing problems. • Trained multiclass machine-learning classifiers (ROOT TMVA) to separate $\Lambda\Lambda$ signal from background. • Built a staged (S0–S3) analysis pipeline with automated systematic variations on the GSI batch farm (Slurm), developed with large language models. • Wrote an event generator and detector-response folding for CBM feasibility studies of $\Lambda\Lambda$ cusps; contributed to the QCD at FAIR white paper. C++ • ROOT • Python • TMVA • Slurm • PLUTO • GEANT • KinFit • Git
Jun–Aug 2019	Summer research project: density-matrix dynamics <i>KTH Royal Institute of Technology</i> Stockholm, Sweden Literature study on the dynamics of the density matrix and quantum entanglement. Supervision: Prof. Erik Aurell

Oct 2018 - 2019 **Research project: flux transfer events**
KTH Royal Institute of Technology, Space and Plasma Physics Stockholm, Sweden
 Statistical study of flux transfer events near Earth's magnetosphere using ESA Cluster spacecraft data, supported by NASA MMS data.
 Supervision: Asst. Prof. Anita Kullen, Prof. Tomas Karlsson

INDUSTRY & ENGINEERING EXPERIENCE

Sep 2018 - 2019 **Electric Propulsion Engineer**
KTH Hyperloop team Stockholm, Sweden
 Linear induction motor for the SpaceX Hyperloop Pod Competition 2019.
 • Designed a double-sided linear induction motor for the pod's propulsion; electromagnetic finite-element simulations in COMSOL and ANSYS Maxwell, with MATLAB.
 COMSOL • ANSYS Maxwell • MATLAB

Aug 2017 - Jul 2018 **Associate Software Engineer**
Robert Bosch Engineering and Business Solutions (RBEI) Bengaluru, India
 • Worked on signal-processing algorithms for automotive radar sensors used in driver-assistance systems, in an industrial software-engineering environment.

Jan-Jul 2017 **Intern, Network Engineering Facility**
AT&T Bengaluru, India
 VPN order-management processes in a network-engineering organisation.

Aug 2016 - Jun 2017 **Hybrid levitation for Hyperloop: engineering project**
BMS College of Engineering Bengaluru, India
 Design of air-bearing levitation combined with linear-induction-motor propulsion for a Hyperloop concept.

Jul 2015 **Intern, Quality Assurance division**
ISRO, Laboratory for Electro-Optics Systems (LEOS) Bengaluru, India
 Circuit analysis of star, Earth and Sun sensors for the ASTROSAT and Chandrayaan-2 missions.

PUBLICATIONS

ARTICLES & PROCEEDINGS

- [1] **G. Appagere** (for the HADES and CBM Collaborations), "Investigating Double-Strange Systems at FAIR: $\Lambda\Lambda$ interactions and neutron star cores", *JPS Conference Proceedings, 15th International Conference on Hypernuclear and Strange Particle Physics (HYP2025), Tokyo (2026)*. **IN PRESS**

COLLABORATION PAPERS & REPORTS

- [2] J. G. Messchendorp, F. Nerling, ..., **G. Appagere**, et al., "**Hadron Physics Opportunities at FAIR**", *arXiv:2512.15986 [hep-ex] (2025)*. **PREPRINT**
 Co-author; contributed the $\Lambda\Lambda$ cusp feasibility study for CBM to the baryon-baryon interactions chapter.

THESES

- [3] **G. Appagere**, " Λ - Λ hyperons: the road to estimate the production cross-section in proton-proton collisions", *Licentiate thesis, Department of Physics, Stockholm University (2024)*.
- [4] **G. Appagere**, "Can FIMPs be Dark Matter Candidates?", *Master's thesis, KTH Royal Institute of Technology (advisor: Prof. R. Mann) (2020)*.

CONFERENCE TALKS

Sep 2026 "Cusp effects and near-threshold structures: $\Lambda\Lambda$ interaction simulation study with CBM"
NSTAR 2026: 15th International Workshop on the Physics of Excited Nucleons, Seville, Spain

May 2026 "Hyperon-hyperon interaction studies close to threshold with CBM at FAIR: a feasibility study with the $\Lambda\Lambda$ production channel"
15th Nordic Meeting on Nuclear Physics, Visby, Sweden

Mar 2026 "Probing hyperon-hyperon interactions with CBM at FAIR"
DPG Spring Meeting 2026, Hadrons and Nuclei section (HK 37.2), Erlangen, Germany **INVITED**

Oct 2025	“Investigating double-hyperon systems using HADES and CBM: insights into Λ - Λ interactions and neutron star cores” <i>Annual Swedish Nuclear Physics and SFAIR Meeting 2025</i> , Chalmers, Göteborg, Sweden CONTRIBUTED
Oct 2025	“Investigating double-strange systems at FAIR: Λ - Λ interactions and neutron star cores” <i>HYP2025: 15th International Conference on Hypernuclear and Strange Particle Physics</i> , Tokyo, Japan CONTRIBUTED
Oct 2024	“ Λ - Λ hyperons: journey towards the understanding of the hyperon–hyperon interactions in neutron stars” <i>Annual Swedish Nuclear Physicists’ and SFAIR Meeting 2024</i> , Lund, Sweden CONTRIBUTED
Oct 2023	“Investigating hyperon–hyperon interaction with the HADES experiment” <i>Annual Swedish Nuclear Physicists’ and SFAIR Meeting 2023</i> , Sweden CONTRIBUTED
Jun 2023	“Understanding the Λ - Λ interaction with the HADES experiment” <i>Fysikdagarna 2023</i> , Stockholm, Sweden CONTRIBUTED
Oct 2022	“ Λ - Λ study using a kinematic fitter in pp reactions at HADES” <i>Annual Swedish Nuclear Physicists’ and SFAIR Meeting 2022</i> , Sweden CONTRIBUTED

COLLABORATION PRESENTATIONS

Feb 2026	Searching for double-strangeness production in pp collisions at $\sqrt{s} = 3.46$ GeV: update on Λ - Λ reconstruction • <i>50th HADES Collaboration Meeting</i> , GSI, Darmstadt, Germany
Oct 2025	From HADES to CBM: investigating Λ - Λ interactions • <i>CBM Collaboration Meeting</i> , online
Apr 2024	Status report on exclusive Λ - Λ reconstruction in pp at 4.5 GeV, towards a cross-section determination • <i>46th HADES Collaboration Meeting</i> , GSI, Darmstadt, Germany
Oct 2023	Status report on exclusive Λ - Λ reconstruction in pp at 4.5 GeV: reconstruction efficiencies compared with simulations • <i>HADES Collaboration Meeting</i> , Stockholm, Sweden
Mar 2023	First Λ - Λ signals from p(4.5 GeV) + p • <i>HADES Collaboration Meeting</i> , GSI, Darmstadt, Germany
Sep 2022	Λ - Λ study with a kinematic fitter in pp at 4.5 GeV: status report • <i>43rd HADES Collaboration Meeting</i> , Coimbra, Portugal

SCHOOLS, WORKSHOPS & EXPERIMENTS

EXPERIMENTS & BEAM TIMES

2025	HADES pion-beam run • GSI, Darmstadt, Germany
Feb 2023	Nuclear-structure experiment JR160 at the JYFL Accelerator Laboratory (neutron-deficient Os isotopes) • Jyväskylä, Finland
Feb 2022	HADES proton beam time (pp at 4.5 GeV): detector shifts • GSI, Darmstadt, Germany

WORKSHOPS & RESEARCH VISITS

Feb 2026	Research visit: Forschungszentrum Jülich (theory) and GSI • Jülich & Darmstadt, Germany
Oct 2025	HAMP2025: Hadron Physics using Antimatter Probes • KEK Tokai, Japan
Jun 2025	QCD at FAIR workshop: contributions to the FAIR hadron-physics white paper • Catania, Italy
2024–2025	Research visits to the GSI hyperon group (CBM feasibility studies) • Darmstadt, Germany

SCHOOLS & COURSES

Sep 2023	HaSP / STRONG-2020 Hadron Spectroscopy Summer School • Salamanca, Spain
Oct 2022	Basic course in radiation safety, Stockholm University • Stockholm, Sweden

TEACHING & SUPERVISION

COURSES

2021–2025	Nuclear Physics (MSc) , FK7051 • HT2021, HT2023, HT2025 Laboratory supervisor: lab introduction lecture, lab compendium, grading of lab reports and oral presentations
2021–2023	Programmering, numeriska metoder och statistik för fysiker , FK4026 • HT2021, HT2022, HT2023 Laboratory assistant: Python labs, numerical-methods part, project grading
2024	Detector Physics (MSc) , FK7056 • HT2024 Laboratory supervisor: muon lifetime and HPGe gamma-spectroscopy labs
2024–2025	Project in Experimental Modern Physics , FK5026 • HT2024, HT2025

- Laboratory supervisor: Compton-scattering lab
- 2025 **Elektromagnetism och vågor**, FK5019 • VT2025
Problem-solving sessions
- 2025 **Vågrörelselära och optik (Waves and Optics)** • HT2025
Exercise-group leader and grading of hand-ins
- SUPERVISION
- 2022 **Bachelor's thesis in Medical Radiation Physics** • VT2022
Co-supervisor: Monte Carlo study of a high-resolution germanium (HPGe) detector at the AlbaNova low-background laboratory
- PEDAGOGICAL TRAINING
- 2021 **Introduction to teaching (3 hp), Stockholm University**
Higher-education teacher training

LEADERSHIP

- 2022-2026 **Chair** (2025–2026), **Vice-Chair** (2023–2025), **Board member** (2022–2023), Fysikum PhD Council, Stockholm University. Lead the department's doctoral-student council: run board and general meetings and represent PhD students in departmental bodies. Contributed to clearer departmental guidelines for changing supervisors and to an improved half-time review process; started PhD-student meetings in the Instrumentation division.
- 2024-2025 **Vice-Chair**, Central Doctoral Student Council (CDR), Stockholm University. University-wide representation of doctoral students; represented CDR in national doctoral-student bodies.
- 2024 **Vice-Chair**, Natural Sciences Faculty Doctoral Student Council (NFR), Stockholm University. Faculty-level doctoral-student representation; frequently meeting secretary.
- 2023-2026 **Doctoral-student representative**, Fysikum Department Board and committees (Research Education, Technical Division, Computer, Work Environment). Department Board member and committee seats on research education, technical-division and computing committees.
- 2025 **Co-organiser**, NFR Debate Forum, Stockholm University. Oxford-style debates for doctoral students across the Faculty of Science.
- 2021-2025 **Lab responsible**, Nuclear physics student laboratory, Fysikum. Responsible for HPGe detectors, scintillators and the radioactive-source inventory; lab rules and safety, a source booking and record system, and procurement of a proportional counter.
- 2024-2025 **Mentee**, Industry mentor programme for doctoral students. Paired with a senior industry mentor to prepare the transition from academia to industry.
- 2015 **Executive Committee Secretary**, IEEE Student Chapter, BMS College of Engineering.

GRANTS & AWARDS

GRANTS & SCHOLARSHIPS

- 2026 **C F Liljevalch J:ors travel grant**, Stockholm University. Conference travel.
- 2025 **Donation scholarship, Faculty of Science**, Stockholm University. Travel to HYP2025, Tokyo.
- 2022 **Birger och Gurli Grundströms forskarstipendiefond i fysik, travel stipend**, Department of Physics, Stockholm University. Participation in an experiment at the JYFL Accelerator Laboratory, Jyväskylä.
- 2022 **Donation scholarship, Faculty of Science (August Emil Wilhelm Smitts stipendie- och understödsstiftelse)**, Stockholm University. Travel to HADES collaboration meetings.

AWARDS

- 2017 **SAE E-Baja 2017: 3rd place in India, design review**, SAE India.
- 2016 **IEEE Darrel Chong Student Activity Award**, IEEE. Special-interest-group activities teaching off-syllabus topics.
- 2015 **SAE Aero Design West 2015: 13th worldwide (team captain)**, SAE International, Van Nuys, CA.

SKILLS

Data analysis, ML & AI	Statistical inference (maximum-likelihood & template fits, least squares, sideband methods) · Uncertainty quantification (systematic variations, correlated averages) · Kinematic (constrained) fitting (vertex and multi-constraint fits) · Machine-learning classification (multiclass classifiers in ROOT TMVA) · AI-assisted development (large language models for building and automating pipelines) · Machine Learning for Physicists and Astronomers (PhD course, 7.5 hp)
Programming & software	Python (NumPy, SciPy, pandas, matplotlib, uproot,iminuit, Jupyter) · C++ & ROOT (analysis frameworks, PyROOT, RooFit) · C · Git / GitLab · Bash & batch scripting · JavaScript, HTML, CSS · LaTeX
Simulation & modelling	Monte Carlo event generation (PLUTO, UrQMD, own generator with final-state interactions) · Detector simulation (GEANT3-based HGeant, CbmRoot) · Finite-element electromagnetics (COMSOL, ANSYS Maxwell) · Mathematica, MATLAB
Detectors & instrumentation	Tracking & timing detectors (drift chambers, straw tubes, RPC, LGAD start detector, time of flight) · Gamma spectroscopy (HPGe detectors, scintillators, neutron activation) · Beam-time operations & online data-quality monitoring · Radiation safety & radioactive-source handling · Automotive radar signal processing
Computing & HPC	HPC batch computing (Slurm on the GSI Virgo cluster) · Containers & software distribution (Singularity/Apptainer, CVMFS) · Large datasets (~155 TB raw data, Lustre, ROOT trees, HDF5)

LANGUAGES

English	Fluent (professional working proficiency)
Swedish	Basic (A2, studying)